

BACKLOG SUBJECT EXAMINATIONS – AUGUST / SEPTEMBER 2023

Program	: B.E. – Information Science and Engineering	Semester	: III
Course Name	: Object Oriented Programming with Java	Max. Marks	: 100
Course Code	: IS34	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT - I

- List any four characteristic features of Java language and describe each one of them in brief. CO1 (06)
 - Write a Java program to print 'n' Fibonacci numbers. Accept the input in command line prompt. Display the output in a well formatted way. CO1 (08)
 - Write a Java program to convert time in seconds to hours, minutes and seconds, and display the output in format HH:MM:SS CO1 (06)
- Describe the following OOP terminologies: CO1 (06)
 - Classes and Instances
 - Command Line Arguments and Variable-Length Arguments.
 - Write a Java program to rearrange all the elements of a given array of integers so that all the odd numbers come before all the even numbers. CO1 (08)
 - Write a Java program to read an integer n and find the number of combinations of a,b,c and d ($0 \leq a,b,c,d \leq 9$) where $(a + b + c + d)$ will be equal to n. CO1 (06)

UNIT - II

- What is the output of the following program? Justify your answer. CO2 (06)

```

class Base
{
    int i;
    private int j;
    void setij(int x, int y)
    {
        i=x;
        j=y;
    }
}
class Derive extends Base
{
    int total;
    void sum()
    {
        total = i + j;
    }
}

public class SubDemo
{
    public static void main(String args[])
    {
        Derive ob= new Derive();
        ob.setij(10,20);
        ob.sum();
        system.out.println(total);
    }
}

```

- b) Compare Method Overriding with Method Overloading. Illustrate the method overriding with an example. CO2 (08)
- c) "One interface can inherit another interface", illustrate the same with code snippet. CO2 (06)
- 4. a) Describe how to pass the arguments to the superclass using code snippet. CO2 (06)
- b) Differentiate between Abstract Class and Interface with code snippet. CO2 (08)
- c) "Package is both a naming and a visibility control mechanism" Justify the statement. CO2 (06)

UNIT - III

- 5. a) Find the type of exceptions in the following code and justify your answer with the cause of error: CO3 (08)
 - i.

```
class ArrayAccess {  
    public static void main(String[] args) {  
        String[] students = {"Shreya", "Amit", null};  
        System.out.println(students[3].length());  
    }  
}
```
 - ii.

```
class FileWrite {  
    public static void main(String[] args) {  
        PrintWriter pw = new PrintWriter("abc.txt");  
        pw.write("print");  
    }  
}
```
 - iii.

```
class Test {  
    public static void main(String[] args) {  
        Thread t = new Thread();  
        t.start();  
        t.sleep(1000);  
    }  
}
```
 - iv.

```
class Test1 {  
    static void m1() {  
        m2();  
    }  
    Static void m2(){  
        m1();  
    }  
    Public static void main(String[] args){  
        m1();  
    }  
}
```
- b) Identify caught or uncaught exceptions in the following: CO3 (06)
 - i. RuntimeException
 - ii. ArithmeticException
 - iii. InterruptedException
 - iv. Error
 - v. NullPointerException
 - vi. FileNotFoundException.
- c) Write Java code to demonstrate try with multiple catch blocks and order of Exception in those catch blocks. CO3 (06)

6. a) List any 5 in-built exception classes and write the code which cause each of those exceptions. CO3 (08)
b) Differentiate between throw and throws clauses with suitable examples. CO3 (06)
c) Write Java code to demonstrate the working of multi-catch block. CO3 (06)

UNIT- IV

7. a) Write a Java program to convert following primitive types to the object of their corresponding wrapper classes using autoboxing mechanism. Also perform the reverse of autoboxing and print their values. CO4 (08)
i) char
ii) boolean
iii) int
iv) double
b) Differentiate between ArrayList and LinkedList classes in Java collections. CO4 (04)
c) Write a Java program to implement HashSet and store unique Objects. CO4 (08)
8. a) Describe the following methods supported by the wrapper classes with their syntax. CO4 (08)
i) typeValue()
ii) valueOf()
iii) toString()
iv) equals()
b) Differentiate between Iterator and Collection in Java programming language. CO4 (04)
c) Write a Java program to create two linked lists which stores integer elements. Use 'for-each' loop to add the two linked lists and display the same. CO4 (08)

UNIT - V

9. a) Consider the following Java code: CO5 (08)
`MyRunnablemr = new MyRunnable(); //MyRunnable implements Runnable`
`Thread t1 = new Thread ();`
`Thread t2 = new Thread(mr);`
Analyze the following method calls and describe their response:
i. `t1.start()`
ii. `t2.start()`
iii. `mr.start()`
iv. `mr.run()`
b) Write the Java code to establish interthread communication using `wait()` and `notify()` methods. CO5 (06)
c) Illustrate the relationship between different states of a thread. CO5 (06)
10. a) Write Java code which demonstrates: CO5 (08)
i. Lambda expression without parameters
ii. Lambda expressions with parameters
iii. Lambda expressions with return statements.
b) Write a program in Java using lambda expression that computes the factorial of a number. CO5 (06)
c) Write a method which has lambda as parameter that removes spaces from the given string. CO5 (06)
